



Linear Models II: Accuracy and Assumptions

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Stat 141 | Fall 2026
Friday, Week 4

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Announcements/Reminders

- I *highly* recommend doing the reading before Monday's class!

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Goals for Today

- Review simple linear regression concepts and R functions
- Discuss model assumptions for linear regression
- Assess accuracy of linear regression models
- Discuss “extrapolation”
- If time - week 4 feedback

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Form of the Model

$$y = f(x) + \epsilon$$

where ϵ represents an error term.

Goal:

- Determine a reasonable form for $f()$. (Ex: Line, curve, ...)
- Estimate $f()$ with $\hat{f}()$ using the data.
- Generate predicted values: $\hat{y} = \hat{f}(x)$.

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Simple Linear Regression Model

$$y = \beta_0 + \beta_1 x + \epsilon$$

Consider this model when:

- Response variable (y): quantitative
- Explanatory variable (x): quantitative
 - Have only ONE explanatory variable.
- AND, $f()$ can be approximated by a line.
- Need to determine the best **estimates** of β_0 and β_1 .

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Distinguishing between the population and the sample

$$y = \beta_0 + \beta_1 x + \epsilon$$

Population model

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Fitted or estimated model

- | | |
|--|---|
| <ul style="list-style-type: none"> • Parameters: <ul style="list-style-type: none"> ▪ Based on the population ▪ Unknown if we don't have data on the whole population ▪ EX: β_0 and β_1, AKA population coefficients | <ul style="list-style-type: none"> • Statistics: <ul style="list-style-type: none"> ▪ Based on the sample data ▪ Known ▪ Usually estimate a population parameter ▪ EX: $\hat{\beta}_0$ and $\hat{\beta}_1$, AKA estimated or "fitted" coefficients |
|--|---|

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Regression goals

We'll use linear regression for 2 tasks (mostly the first):

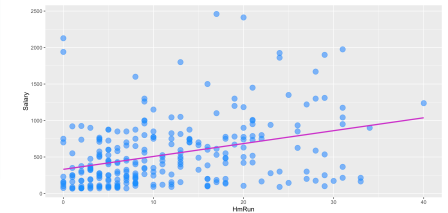
- **Explaining** relationships between variables in a data set so that we can **infer** relationships in the population
 - Ex. We observe that a sample of Reed students who sleep more have higher test scores, infer about the population **in general**
 - We focus most on **coefficients**
- **Predicting** values of the response variable based on values of the explanatory variable
 - Ex. Using data from 1960-2015, we predict the atmosphere will contain 410 ppm CO2 in 2025
 - We focus most on **fitted values** and **residuals**

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Our Modeling Goal

Recall our modeling goal: **predict MLB player salary in 1986/1987 by using the HmRun variable.**

```
1 library(ISLR)
2 data(Hitters)
3 hitters <- Hitters %>% filter(!is.na(Salary), !is.na(HmRun))
4
5 ggplot(data = hitters,
6       mapping = aes(x = HmRun,
7                     y = Salary)) +
8   geom_point(alpha = 0.6, size = 4,
9             color = "dodgerblue") +
10  geom_smooth(method = "lm", se = FALSE,
11             color = "magenta3")
```

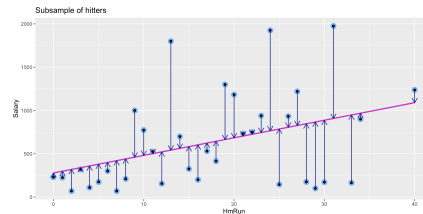


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Method of Least Squares

- Want residuals to be small.
- Minimize a function of the residuals.
- Minimize:

$$\sum_{i=1}^n e_i^2$$



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Method of Least Squares

After minimizing the sum of squared residuals, you get the following equations:

Get the following equations:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

where

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i \quad \text{and} \quad \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

- Note: A large slope does **not necessarily** indicate strong correlation, and a small slope does **not necessarily** indicate lack of correlation.

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Method of Least Squares

Then we can estimate the whole function with:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$$

Called the **least squares line**, **regression line**, **fitted line/model**, or the **line of best fit**.

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Constructing the Simple Linear Regression Model in R

We can use the `lm()` function to construct the simple linear regression model in R and the `get_regression_table()` function from `moderndive` to interpret it.

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
2 get_regression_table(mod)

# A tibble: 2 x 7
  term      estimate std_error statistic p_value lower_ci upper_ci
<chr>    <dbl>      <dbl>    <dbl>   <dbl>   <dbl>    <dbl>
1 intercept 330.59      43.551    7.591     0    244.84   416.35
2 HmRun      17.671      2.995     5.9      0     11.773   23.569
```

Q: What is the fitted model form?

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \times x_{\text{HmRun}}$$

$$= 330.59 + 17.67 \times x_{\text{HmRun}}$$

Q: How do we interpret the coefficients?

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Coefficient Interpretation

$$\begin{aligned}\hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 \times x_{\text{HmRun}} \\ &= 330.59 + 17.67 \times x_{\text{HmRun}}\end{aligned}$$

We need to be precise and careful when interpreting estimated coefficients!

- **Intercept:** We expect/predict y to be $\hat{\beta}_0$ on average when $x = 0$.
- **Slope:** For a one-unit increase in x , we expect/predict y to change by $\hat{\beta}_1$ units on average.

This template is non-specific to the context of our model, but when we are actually interpreting coefficients, we always need to insert the context

- Use variable names instead of x and y
- Insert estimated values for $\hat{\beta}_0, \hat{\beta}_1$
- Explain what “ $x = 0$ ” and a “one-unit increase in x ” corresponds to, if you can do so briefly

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Coefficient Interpretation: In Context

$$\begin{aligned}\hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 \times x_{\text{HmRun}} \\ &= 330.59 + 17.67 \times x_{\text{HmRun}}\end{aligned}$$

- **Intercept:** We expect/predict a player's **Salary** to be \$330,000 on average when they've hit 0 home runs.
- **Slope:** For **each additional home run** (or one-unit increase in **HmRun**) that a player hits, we expect/predict their **Salary** to increase by \$17,670 on average.

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Prediction, Interpolation, and Extrapolation

```
1 new_cases <- data.frame(HmRun = c(10, 32, 60)) # Must have same explanatory variable name as mod!
2 predict(mod, newdata = new_cases)

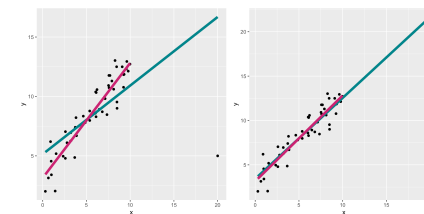
      1      2      3
507.3030 896.0638 1390.8503
```

- We didn't have any players in our sample with 32 home runs. Can we still make this prediction?
 - Called **interpolation**
- We didn't have any players in our sample with 60 home runs. Can we still make this prediction?
 - Called **extrapolation**

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Cautions

- Careful to only predict values within the range of x values in the sample.
- Make sure to investigate **outliers**: observations that fall far from the cloud of points.
 - High leverage points: extreme *predictor* value



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A closer look at our model

```
1 get_regression_table(mod)
# A tibble: 2 × 7
  term      estimate std_error statistic p_value lower_ci upper_ci
  <chr>      <dbl>    <dbl>    <dbl>   <dbl>   <dbl>   <dbl>
1 intercept 330.59     43.551     7.591     0 244.84 416.35
2 HmRun      17.671     2.995     5.9      0 11.773 23.569
```

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 \times x_{\text{HmRun}} \\ = 330.59 + 17.67 \times x_{\text{HmRun}}$$

What assumptions have we made?

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Linear Regression Assumptions

We can *always* find the line of best fit to explore data, but...

To make accurate predictions or inferences, certain conditions should be met.

To responsibly use linear regression tools for prediction or inference, we require:

- Linearity:** The relationship between explanatory and response variables must be approximately linear
 - Check using scatterplot of data, or **residual plot**
- Independence:** The observations should be independent of one another.
 - Can check by considering data context, and
 - sometimes by looking at scatterplots/**residual plots**
- Normality:** The distribution of residuals should be *approximately* bell-shaped, unimodal, symmetric, and centered at 0 at every “slice” of the explanatory variable
 - Simple check: look at **histogram of residuals**
 - Better to use a **Q-Q plot**
- Equal Variability:** Variance of residuals should be roughly constant across data set. Also called “homoscedasticity”. Models that violate this assumption are sometimes called “heteroscedastic”
 - Check using **residual plot**.

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A cute way to remember this: “LINE”

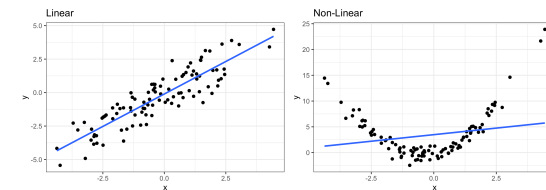
- L**inearity
- I**ndependence
- N**ormality
- E**qual Variability

- We assess these using **diagnostic plots** (y vs x scatterplot, residual plot, residual histogram, q-q plot)
- Later in the course, we’ll learn how **I**, **N**, and **E** are required for **inference**

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Assessing conditions: Linearity

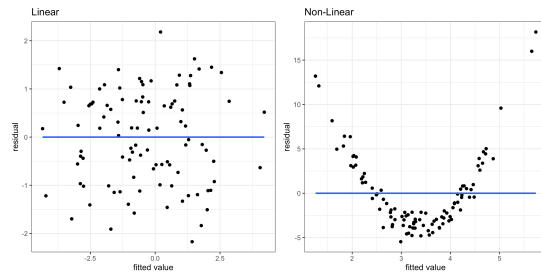
Linearity: The relationship between explanatory and response variables must be approximately linear



- Points should be evenly distributed above and below the regression line **at each “slice” of x**
- If data is non-linear:
 - Slope does not adequately describe relationship and predictions can be very inaccurate
 - More advanced modeling techniques should be used instead (take STAT 243)

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Assessing conditions: Linearity

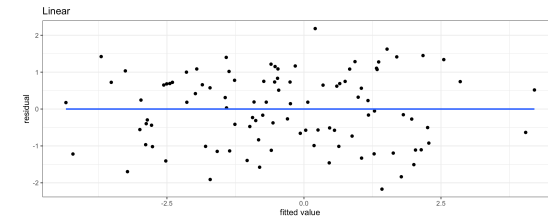


- Residual plots can also be useful for checking linearity
- These plot model residuals against the fitted values
- We don't want to see any trends here

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Assessing conditions: Linearity (Code Example)

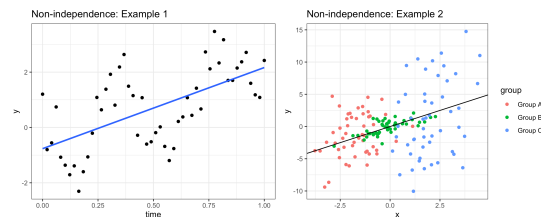
```
1 library(moderndivde)
2
3 # Creating the model
4 lm1 <- lm(data = my_df, y1 ~ x)
5
6 # *** Pulling out model inputs, residuals, and fitted values ***
7 res1 <- get_regression_points(lm1)
8
9 # Plotting residuals vs fitted values
10 (g1 <- ggplot(res1, aes(x = y1_hat, y = residual)) +
11   geom_point() +
12   theme_bw() +
13   geom_smooth(method = "lm", se = F) + # Helps check for any trend - want a flat line here
14   labs(x = "fitted value", y = "residual", title = "Linear"))
```



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Assessing conditions: Independence

Independence: The observations should be independent of one another

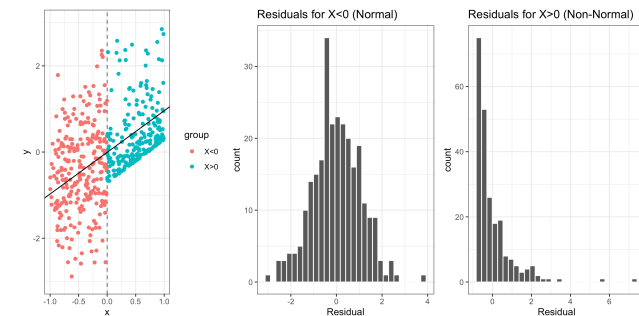


- If observations are not independent, inference about the population may be misleading.
- Think about independence between observations (rows in the data frame), not necessarily between variables (columns in the data frame).
- We usually assess independence qualitatively: Are observations inherently connected?

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Assessing conditions: Normality (of residuals)

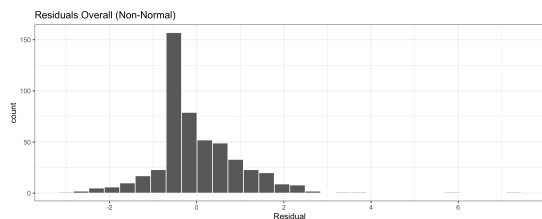
Normality: The distribution of residuals should be bell-shaped, unimodal, symmetric, and centered at 0 at every “slice” of the explanatory variable



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Assessing conditions: Normality (of residuals)

Normality: The distribution of residuals should be bell-shaped, unimodal, symmetric, and centered at 0 at every “slice” of the explanatory variable



If residuals are non-Normal...

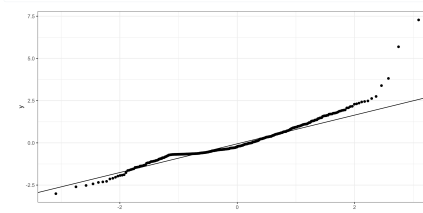
- Some predictions can be very inaccurate
- Inference about the population may be misleading

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Assessing conditions: Normality (of residuals) (QQ-Plot)

Normality: The distribution of residuals should be “Normal”

```
1 # Creating the model
2 lm2 <- lm(data = my_df, y ~ x)
3
4 # *** Pulling out model inputs, residuals, and fitted values
5 res2 <- get_regression_points(lm2)
6
7 # Making a Q-Q plot: new aes() argument!!
8 ggplot(res2, aes(sample = residual)) +
9   geom_qq() +
10  geom_qq_line() +
11  theme_bw()
```

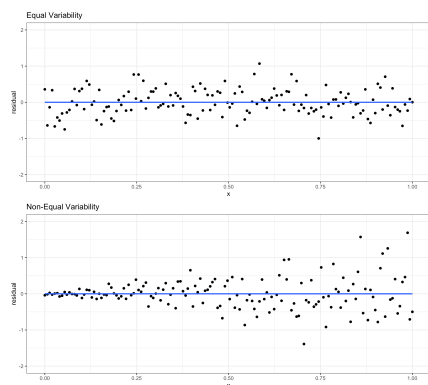


- “Q-Q” plots make it even easier to check Normality - we’ll learn some of the theory behind this later.
- We want points to fall as close to the line as possible.

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Assessing conditions: Equal variability (of residuals)

Equal Variability: Variance of residuals should be approximately constant across the data

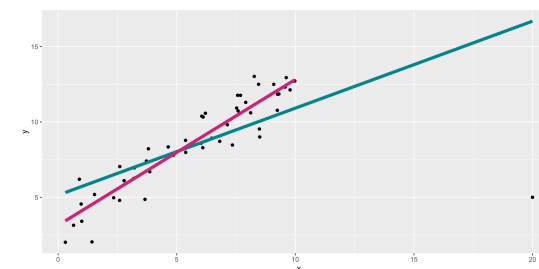


- **Residual plots** are also very useful for this
- If equal variability isn’t met:
 - Inference about the population may be misleading
 - (Potential) outliers in high-variability range are more influential

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Another example: high leverage points

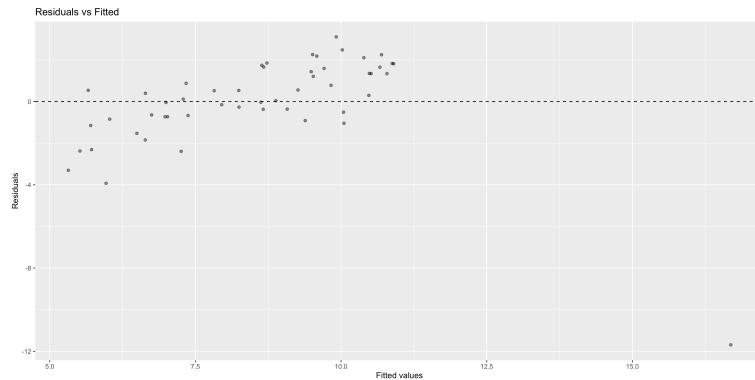
Remember the outlier example:



What do diagnostics look like when we fit the teal model?

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Diagnosing the model



- In this case, can already see the outlier in the residual vs fitted plot.

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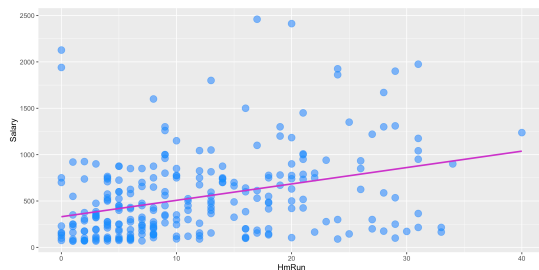
What to do with high leverage points?

- Depends on **context**
- It may be reasonable to remove them and refit the model
 - If we believe the data is the result of an error
 - If we are most focused on **prediction** for non-outliers, or if our population of interest does not include the outlier
- But we shouldn't remove outliers just because we don't like the results that they produce
 - Especially if our goal is **descriptive** or **explanatory**, and the outlier represents a valid data point

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Recall our **hitters** simple linear regression model

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
```



Let's check if this meets the **LINE** assumptions

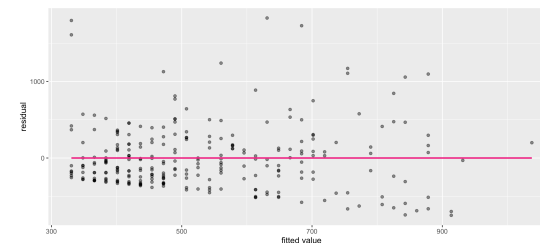
Q: First, what does this graph tell us about **Linearity**?

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Residual vs. fitted plot:

Remember to use `get_regression_points(mod)` (not `get_regression_TABLE`) to get residuals and fitted values from a model.

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
2 res <- get_regression_points(mod)
3
4 ggplot(res, aes(x = Salary_hat, y = residual)) +
5   geom_point(alpha = 0.5) +
6   geom_smooth(method = "lm", se = FALSE, color = "deeppink2") +
7   labs(x = "fitted value", y = "residual")
```



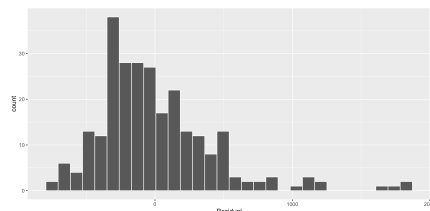
- Linearity:
- (`geom_smooth()` helps check this again)
- Independence: **?**
- Equal Variability:

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Checking normality: residual histogram

Remember to use `get_regression_points(mod)` (not `get_regression_TABLE`) to get residuals and fitted values from a model.

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
2 res <- get_regression_points(mod)
3
4 ggplot(res, aes(x = residual)) +
5   geom_histogram(bins = 30, color = "white") +
6   labs(x = "Residual")
```



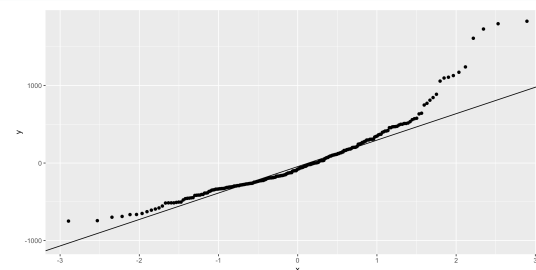
• Normality: 🤔

- Not horrible but not great
- We'll look at a Q-Q plot to further assess!

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Checking normality: Q-Q plot

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
2 res <- get_regression_points(mod)
3
4 ggplot(res, aes(sample = residual)) +
5   geom_qq() +
6   geom_qq_line()
```



• Normality: 🚩

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Suppose LINE assumptions are met. Is the model **good**?

Coefficient of Determination, or R^2

A common measure of the **strength** of a linear model is the **coefficient of determination** R^2 , (aka "R-squared").

- R^2 is always between 0 and 1 (for the linear models we discuss)
- It measures the **proportion of variation in the response variable y that is explained by the linear model**

$$R^2 = \frac{\overbrace{s_y^2 - s_e^2}^{\text{Variation in Y explained by X}}}{\underbrace{s_y^2}_{\text{Variation in y}}} \quad (\text{Reminder: Variance} = (\text{Standard Deviation})^2)$$

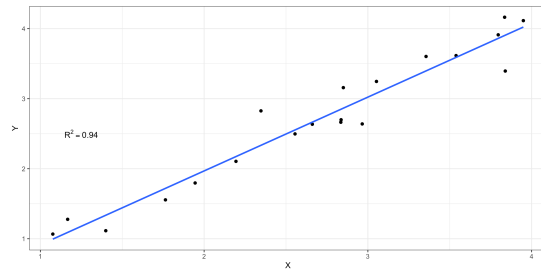
- s_y^2 = **Total Variation in y**
- s_e^2 = Variation in the residual (e) = Unexplained Variation in y
- $s_y^2 - s_e^2$ = (Total Variation in y) - (Unexplained Variation in y) = **Variation in y explained by x**

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Example of High R^2

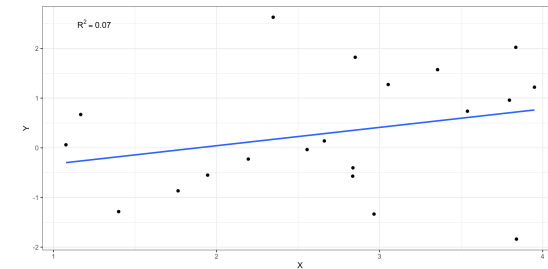
If $R^2 \approx 1$: nearly all the variability in response is **explained by** variability in the explanatory variable.



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Example of Low R^2 (and code to get it)

If $R^2 \approx 0$: almost none of the variability in response is **explained by** variability in the explanatory variable.



Yet another **moderndive** function can help us calculate R^2 from a fitted model:

```
1 mod <- lm(Salary ~ HmRun, data = hitters)
2 get_regression_summaries(mod)$r_squared
[1] 0.118
```

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Recap

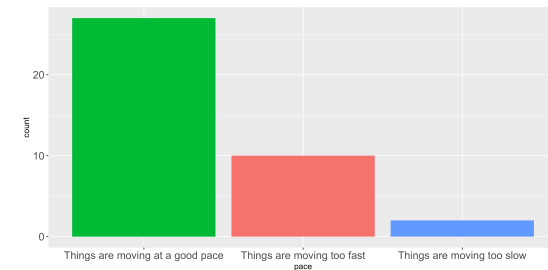
- Linear models must meet the four **LINE** assumptions to have reliable output
 - We can check these using a combination of plots and contextual knowledge about the data
- We should be careful about **outliers** and **extrapolation** when fitting our linear models and using them to predict for new observations
- The coefficient of determination, R^2 , gives a measure of how well a regression line fits our data
- Key functions:
 - lm()** to fit a model
 - get_regression_table()** to calculate coefficients from a model
 - get_regression_points()** to grab residuals/fitted values from a model
 - get_regression_summaries** to calculate R^2 from a model
 - predict()** to make predictions for new x values from a model

Next Time

- Regression with one categorical variable

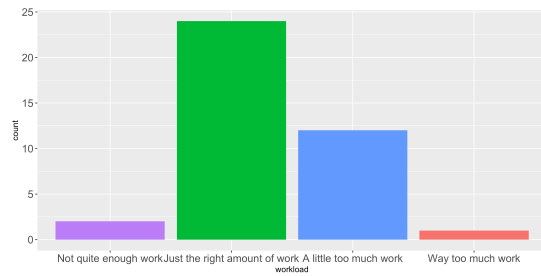
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4 week feedback - pace



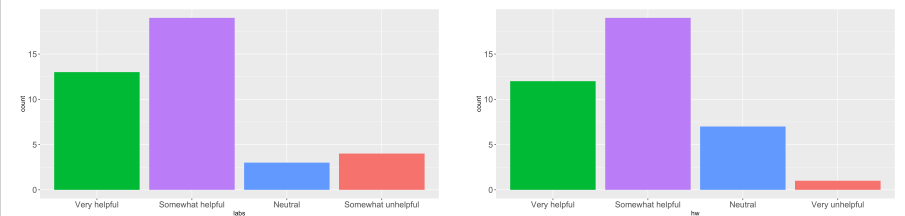
40

4 week feedback (workload)



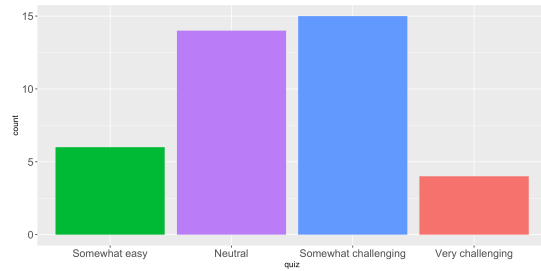
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4 week feedback (labs and HW)



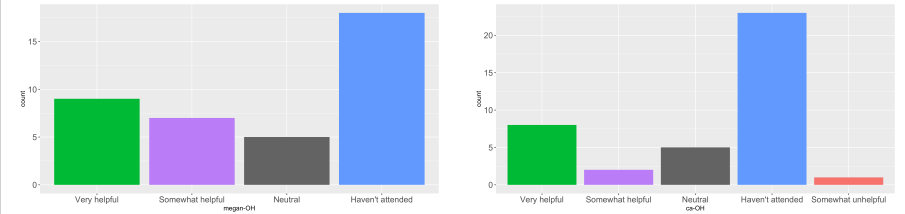
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4 week feedback (quiz)



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4 week feedback (OH)



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